1 Deriving the 2D labelled functional equation

This lengthy inclusion-exclusion-type derivation yields the following generating function equation

$$\begin{aligned}
Q = 1 &+ tSQ - t\overline{x_2}S_2^{-1}Q_2^0 - t\overline{x_1}S_1^{-1}Q_1^0 + t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0} \\
&+ t(a_1-1)(S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0}) \\
&+ t(a_2-1)(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0}) \\
&+ t(a_1-1)(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})
\end{aligned}$$

We can extract various coefficients from this equation

$$\begin{aligned}
Q_1^0 = 1 &+ ta_1(S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0} \\
&+ ta_1(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})
\end{aligned}$$

$$\begin{aligned}
Q_2^0 = 1 &+ ta_2(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x}_1S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0} \\
&+ ta_2(a_1-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})
\end{aligned}$$

$$Q_{1,2}^{0,0} = 1 + ta_1a_2(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})$$

There is a somewhat hidden pattern in these equations. We let

$$A = SQ - \overline{x_2}S_2^{-1}Q_2^0 - \overline{x_1}S_1^{-1}Q_1^0 + \overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}$$

Then extracting coefficients yields

$$\begin{aligned}
A_1^0 &= S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0} \\
A_2^0 &= S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0} \\
A_{1,2}^{0,0} &= S_{1,2}^{0,-1}Q_{1,2}^{0,1}S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}
\end{aligned}$$

and this allow us to re-write the above equations in a much neater format. First let $b_i = a_1 - 1$ (this will prove very beneficial later). The equations read in a very neat way as

$$\begin{aligned}
Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\
Q_1^0 &= 1 + ta_1A_1^0 + ta_1b_2A_{1,2}^{0,0} \\
Q_2^0 &= 1 + ta_2A_2^0 + ta_2b_1A_{1,2}^{0,0} \\
Q_{1,2}^{0,0} &= 1 + ta_1a_2A_{1,2}^{0,0}
\end{aligned}$$

If none of the terms are 0 (as we'll see later) we can back-substitute the A's in order to obtain an equation relating Q, Q_1^0, Q_2^0 and $Q_{1,2}^{0,0}$. First

$$A_{1,2}^{0,0} = \frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}$$

$$A_1^0 = \frac{Q_1^0 - 1}{ta_1} - (a_2 - 1)A_{1,2}^{0,0} = \frac{Q_1^0 - 1}{ta_1} - (a_2 - 1)\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}$$

$$A_2^0 = \frac{Q_2^0 - 1}{ta_2} - (a_1 - 1)A_{1,2}^{0,0} = \frac{Q_2^0 - 1}{ta_2} - (a_1 - 1)\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}$$

We can substitute these expressions into the initial equation along with $c_i = b_i/a_i = 1 - \overline{a_i}$

$$\begin{aligned} Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\ &= 1 + tA + tb_1\left(\frac{Q_1^0 - 1}{ta_1} - b_2\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}\right) + tb_2\left(\frac{Q_2^0 - 1}{ta_2} - b_1\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}\right) + tb_1b_2\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2} \\ &= 1 + tA + c_1(Q_1^0 - 1) + c_2(Q_2^0 - 1) - c_1c_2(Q_{1,2}^{0,0} - 1) \end{aligned}$$

Expanding the definition of A and re-arranging gives the following expression

$$Q(1 - tS) - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

This suggests the following notation

$$K_I = c_I - t\overline{x_I}S_I^{-\mathbb{I}}$$

where as in multi-index notation $c_I = \prod_{i\in I} c_i$, $x_I = \prod_{i\in I} x_i$ and $\mathbb{I}$ has $|I|$ ones. When $I = \varnothing$ the products will be empty and equal to 1. This allows us to re-write the above equation in the neater form

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{12}Q_{12}^{00} = \overline{a_1a_2}$$

This equation will be the first step in further analysis of the problem. Before we go on though, we need to say a few words about the case when one of the A's is 0. If $A_1^0 = 0$ or $A_2^0 = 0$, this also means $A_{12}^{00} = 0$. This in turn makes the respective Q's equal to 1. For example, if ${}_1^0 = 0$ then $Q_1^0 = 1$ and $Q_{12}^{00} = 1$ as well. This also leads the respective $S_1^{-1} = 0$ and $S_{1,2}^{-1,-1} = 0$ leaving us with a simpler master equation. But more on that later.

2 Generalization to d dimensions.

The overall notation for the problem so far has been suggestive in that it lends itself to an easy generalization. First of all, in d dimensions, for a step Laurent polynomial S the function A satisfies the following relation

$$A = \sum_{J\subseteq[d]} (-1)^{|J|}\overline{x_J}S_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$$

and the unlabelled problem has an equation that reads as

$$Q = 1 + tA. \tag{1}$$

Then we put a label a_i for each coordinate $1 \leqslant i \leqslant d$ that is equal to zero in a random walk. This gives the equation

$$Q = 1 + t \sum_{I \subseteq [d]} b_I A_I^{\mathbb{O}} \tag{2}$$

We can write out the coefficient extraction explicitly as

$$Q = 1 + t \sum_{\substack{I,J \subseteq [d] \\ I \cap J = \varnothing}} b_I \overline{x_J} \sum_{K \subseteq I} \left(S_{J,K}^{-\mathbb{I},-\mathbb{I}} Q_{J,K}^{\mathbb{O},-\mathbb{I}} + S_{J,K}^{-\mathbb{I},\mathbb{O}} Q_{J,K}^{\mathbb{O},\mathbb{O}} \right)$$

Next, we perform coefficient extractions on equation (2) to obtain the following formula

$$Q_J^{\mathbb{O}} = 1 + t \sum_{I \subseteq [d]} b_I A_{I \cup J}^{\mathbb{O}} \tag{3}$$

$$= 1 + t a_J \sum_{I \subseteq [d] \backslash J} b_I A_{I,J}^{\mathbb{O},\mathbb{O}} \tag{4}$$

even though the second equation takes a little bit of work. Before we continue we have to make sure we know what to do if any of the A's are equal to zero. The following lemma characterizes this situation

Lemma 1. $A_I^{\mathbb{O}} \equiv 0$ if and only if $S_{J,I}^{-\mathbb{I},P} \equiv 0$ for all $J \subseteq [d] \backslash I$ and $P \in \{-1, 0\}^{|I|}$.

alternatively, since $S_{J_1,I}^{-\mathbb{I},P} \equiv 0 \Rightarrow S_{J_2,I}^{-\mathbb{I},P} \equiv 0$ for $J_1 \subseteq J_2$ based on simple extraction, then it only suffices to look at S_I^P Therefore we can re-write the lemma as

Lemma 2. $A_I^{\mathbb{O}} \equiv 0$ if and only if $S_I^P \equiv 0$ for all $P \in \{-1, 0\}^{|I|}$. In particular $A_I^{\mathbb{O}} \equiv 0 \Rightarrow S_I^{-\mathbb{I}} \equiv 0$.

The second part of the lemma guarantees that if an A is zero then the corresponding S will be zero in the specific term K in the master equation. The next part takes some effort but we can use backward induction on $|I|$ to show that

$$A_I^{\mathbb{O}} = \frac{(-1)^{|I|}}{t b_I} \sum_{I \subseteq J \subseteq [d]} (-1)^{|J|} c_J (Q_J^{\mathbb{O}} - 1) \tag{5}$$

Substituting these terms back into equation (2) yields the following calculation

$$Q = 1 + tA + t \sum_{\varnothing \neq I \subseteq [d]} b_I \frac{(-1)^{|I|}}{t b_I} \sum_{I \subseteq J \subseteq [d]} (-1)^{|J|} c_J (Q_J^{\mathbb{O}} - 1) \tag{6}$$

$$= 1 + tA + \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} \sum_{I \subseteq J \subseteq [d]} (-1)^{|J|} c_J (Q_J^{\mathbb{O}} - 1) \tag{7}$$

For each I we are taking all sets inbetween I and $[d]$ for the latter sum. How many times do we see every subset J? We see it once for every subset I. There are $2^{|J|}-1$ such subsets because $I \neq \varnothing$. Since J is non-empty, the number is odd so we have

$$Q = 1 + tA + \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|+1} c_J (Q_J^{\mathbb{O}} - 1) \tag{8}$$

$$= 1 + t \sum_{J \subseteq [d]} (-1)^{|J|} \overline{x_J} S_J^{-\mathbb{I}} Q_J^{\mathbb{O}} + \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|+1} c_J (Q_J^{\mathbb{O}} - 1) \tag{9}$$

$$= 1 + \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|} c_J + tS - \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|} (c_J - t\overline{x_J} S_J^{-\mathbb{I}}) Q_J^{\mathbb{O}} \tag{10}$$

The sum of c_J's combines with 1 to create $\overline{a_{[d]}}$. Rerranging everything else to the LHS leaves us with the d-dimensional master equation

Theorem 3. *Given a step set in $d \geqslant 1$ dimensions with Laurent polynomial S, the labelled generating function satisfies the following functional equation*

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = \overline{a_{[d]}}$$

or if we let $K_I = c_I - t\overline{x_I} S_I^{-\mathbb{I}}$ we have

$$\sum_{I \subseteq [d]} (-1)^{|I|} K_I Q_I^{\mathbb{O}} = \overline{a_{[d]}}.$$

We also have a corollary that takes care of all the cases where $S_I^{-\mathbb{I}} = 0$. We have a modified formula

Corollary 4. For everything as in the theorem we have

$$\sum_{\substack{I \subseteq [d] \\ A_I^{\mathbb{O}} \neq 0}} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = f(a_1, a_2, \ldots, a_d)$$

where

$$f(a_1, a_2, \ldots, a_d) = \overline{a_{[d]}} - \sum_{\substack{I \subseteq [d] \\ A_I^{\mathbb{O}} \neq 0}} (-1)^{|I|} c_I$$

which could easily be simplified further, now that I think about it.

3 Further analysis of the 2D equation

We can perform coefficient extractions of the master equation to get the following working forms

$$(1-tS)Q-(c_1-t\overline{x_1}S_1^{-1})Q_1^0-(c_2-t\overline{x_2}S_2^{-1})Q_2^0+(c_1c_2-t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0}=\overline{a_1a_2} \quad (11)$$

$$(1-ta_1S_1^0)Q_1^0-(c_2-ta_1\overline{x_2}S_{1,2}^{0,-1})Q_{1,2}^{0,0}-ta_1(S_1^{-1}Q_1^1-\overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0})=\overline{a_2} \quad (12)$$

$$(1-ta_2S_2^0)Q_2^0-(c_1-ta_2\overline{x_1}S_{1,2}^{-1,0})Q_{1,2}^{0,0}-ta_2(S_2^{-1}Q_2^1-\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1})=\overline{a_1} \quad (13)$$

$$Q_{1,2}^{0,0}-ta_1a_2(S_{1,2}^{-1,-1}Q_{1,2}^{1,1}+S_{1,2}^{-1,0}Q_{1,2}^{1,0}+S_{1,2}^{0,-1}Q_{1,2}^{0,-1})=1 \quad (14)$$

There might be some operator form that will allow us to write these out exactly.

4 Diagonal steps: Deriving the Extraction-Ready Functional Equations

Let's take the case of the diagonal steps with a finite group G of order 4. Let $\bullet : x_1 \mapsto \overline{x_1}$ and $* : x_2 \mapsto \overline{x_2}$. We have

$$K^\bullet = K^* = K, \quad K_2^\bullet = K_2, \quad K_1^* = K_1$$

while the others change. We can also solve when $K = 0$ for x_1. There are two roots

$$\sigma_1 = \frac{1-\sqrt{1-4t^2(S_1^{-1})^2}}{2tS_1^{-1}}, \quad \overline{\sigma_1} = \frac{1+\sqrt{1-4t^2(S_1^{-1})^2}}{2tS_1^{-1}}$$

with the extra property that

$$\sigma_1 + \overline{\sigma_1} = \frac{1}{tS_1^{-1}}.$$

We can take the master equation and apply $*$ to it

$$-KQ + K_1Q_1^0 + K_2Q_2^0 - K_{1,2}Q_{1,2}^{0,0} + \alpha = 0$$
$$-KQ^* + K_1Q_1^{0,*} + K_2^*Q_2^0 - K_{1,2}^*Q_{1,2}^{0,0} + \alpha = 0$$

From here on we will input $x_1 = \sigma_1$ to rid ourselves of the kernel K. There will be an overloading of notation

$$K_1Q_1^0 + K_2Q_2^0 - K_{1,2}Q_{1,2}^{0,0} + \alpha = 0$$
$$K_1Q_1^{0,*} + K_2^*Q_2^0 - K_{1,2}^*Q_{1,2}^{0,0} + \alpha = 0$$

We can multiply both equation by something to clear Q_2^0 because it involves σ_1 while the other term involves only x_2. We have

$$K_1K_2^*Q_1^0 + K_2^*K_2Q_2^0 - K_2^*K_{1,2}Q_{1,2}^{0,0} + \alpha K_2^* = 0$$
$$K_1K_2Q_1^{0,*} + K_2K_2^*Q_2^0 - K_2K_{1,2}^*Q_{1,2}^{0,0} + \alpha K_2 = 0$$

We subtract these two equations to obtain

$$K_1(K_2^*Q_1 - K_2Q_1^*) - (K_2^*K_{1,2} - K_2K_{1,2}^*)Q_{1,2} + \alpha(K_2^* - K_2) = 0$$
$$K_2^*Q_1 - K_2Q_1^* - \frac{K_2^*K_{1,2} - K_2K_{1,2}^*}{K_1}Q_{1,2} + \alpha\frac{K_2^* - K_2}{K_1} = 0$$

First we have

$$\begin{aligned} K_2^* - K_2 &= t(\overline{x_2} - x_2)S_2^{-1}(\sigma_1) \\ &= t(\overline{x_2} - x_2)(\overline{\sigma_1} + \sigma_1) \\ &= \frac{\overline{x_2} - x_2}{\overline{x_2} + x_2} \end{aligned}$$

We have

$$\begin{aligned} K_2^*K_{1,2} - K_2K_{1,2}^* &= \left(c_2 - tx_2S_2^{-1}\right)\left(c_1c_2 - t\overline{\sigma_1 x_2}\right) - \left(c_2 - t\overline{x_2}S_2^{-1}\right)\left(c_1c_2 - t\overline{\sigma_1}x_2\right) \\ &= c_1c_2t(\overline{x_2} - x_2)S_2^{-1} - c_2t\overline{\sigma_1}(\overline{x_2} - x_2) \\ &= c_2t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) \end{aligned}$$

If we let $v = \overline{x_2} + x_2$ we find

$$\sigma_1 = \frac{1 - \sqrt{1 - 4t^2v^2}}{2tv}, \quad \overline{\sigma_1} = \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv}$$

We have

$$\begin{aligned} (c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) &= \frac{c1}{tv} - \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \\ &= \frac{1}{\overline{x_2} + x_2}\left(-\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2v^2}}{2}\right) \end{aligned}$$

Also

$$\begin{aligned} K_1 &= c_1 - tv\overline{\sigma_1} \\ &= c_1 - tv\frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \\ &= 1 - \overline{a_1} - \frac{1 + \sqrt{1 - 4t^2v^2}}{2} \\ &= -\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2v^2}}{2} \end{aligned}$$

Thus we finally have

$$\begin{aligned} \frac{1}{K_1}(K_2^*K_{1,2} - K_2K_{1,2}^*) &= -\frac{c_2t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1})}{c_1 - t\overline{\sigma_1}v} \\ &= -\frac{c_2t(\overline{x_2} - x_2)(-1)}{tv} \\ &= \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2} \end{aligned}$$

Thus the equation will now read as

$$K_2^* Q_1 - K_2 Q_1^* - \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2} Q_{1,2} + \alpha \frac{\overline{x_2} - x_2}{(\overline{x_2} + x_2) K_1} = 0$$

We just need to take care of the remaining terms. We'll do this by multiplying by $\overline{x_2} + x_2$. We have

$$\begin{aligned} R^* &= (\overline{x_2} + x_2) K_2^* = c_2(\overline{x_2} + x_2) - x_2 = c_2 \overline{x_2} - \overline{a_2} x_2 \\ R &= (\overline{x_2} + x_2) K_2 = c_2(\overline{x_2} + x_2) - \overline{x_2} = c_2 x_2 - \overline{a_2 x_2} \end{aligned}$$

We have a couple of relations here

$$\begin{aligned} R^* - R &= v(K_2^* - K_2) = \overline{x_2} - x_2 \\ \overline{x_2} R^* - x_2 R &= \overline{x_2} c_2 v - 1 - x_2 c_2 v + 1 \\ &= c_2 v(\overline{x_2} - x_2) \end{aligned}$$

The equation becomes

$$(c_2 \overline{x_2} - \overline{a_2} x_2) Q_1^0(x_2) + (\overline{a_2 x_2} - c_2 x_2) Q_1^0(\overline{x_2}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + (\overline{x_2} - x_2) \overline{K}_1 \alpha = 0$$

and it's ready for extraction. Let $\delta_0 = \overline{a_1}(\overline{x_2} - x_2)\overline{K}_1$. We re-write the equation as

$$R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2} \delta_0 = 0$$

We'll use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x_1} S_1^{-1}) Q_1^0 - (c_2 - t\overline{x_2} S_2^{-1}) Q_2^0 + (c_1 c_2 - t\overline{x_1 x_2} S_{1,2}^{-1,-1}) Q_{1,2}^{0,0} = \overline{a_1 a_2}$$

to further our work here. For diagonal steps it is

$$(1 - tS)Q - (c_1 - tv\overline{x_1}) Q_1^0 - (c_2 - t\overline{x_2} S_2^{-1}) Q_2^0 + (c_1 c_2 - t\overline{x_1 x_2}) Q_{1,2}^{0,0} = \overline{a_1 a_2}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0 = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1$$

We can extract higher coefficients too. Let $k \geqslant 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned} 0 &= Q_1^k - tS_1^{-1} Q_1^{k+1} - tS_1^0 Q_1^k - tS_1^1 Q_1^{k-1} - c_2 Q_{1,2}^{k,0} + t\overline{x_2}(S_{1,2}^{-1,-1} Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1} Q_{1,2}^{k,0} + S_{1,2}^{1,-1} Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tv Q_1^{k+1} - tv Q_1^{k-1} - c_2 Q_{1,2}^{k,0} + t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0}) \end{aligned}$$

We can re-write the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2 Q_{1,2}^{k,0} - t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})$$

We'll use induction to prove a nice formula but for that we'll first need to establish another base case because of this splitting of cases for the coefficient extraction. We have

$$\begin{aligned} Q_1^0 &= \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1 \\ Q_1^{0,*} &= \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*} \end{aligned}$$

Combining them with equation (4) gives

$$\begin{aligned} 0 &= R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= R^*(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1) - R(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + (R^* - R)(\overline{a_2} + c_2 Q_{1,2}^{0,0}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}(\overline{x_2} - x_2) + \overline{a_2}\delta_0 \\ &= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}((\overline{x_2} - x_2) + \delta_0) \end{aligned}$$

So we can write the equation

$$R^* Q_1^1 - R Q_1^{1,*} + \overline{a_2}\delta_1 = 0$$

where

$$\delta_1 = \frac{(\overline{x_2} - x_2) + \delta_0}{a_1 tv}$$

This equation together with equation (4) form the base cases for the following lemma

Lemma 5. Let $m \geqslant 0$ be an integer. Then

$$R^* Q_1^m - R Q_1^{m,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0 \tag{15}$$

where δ_m is not a function of a_2.

Proof. We'll show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have any walks end on coordinates $(1, 0)$. Now we assume that the formula is true for all cases up to and including m. Then we use formula (4)

$$\begin{aligned} & R^* Q_1^m - R Q_1^{m,*} \\ =& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + tv(R^* Q_1^{m-1} - R Q_1^{m-1,*}) \\ & + c_2(R^* - R) Q_{1,2}^{m,0} - t(\overline{x_2} R^* - x_2 R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\ =& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + tv(c_2(\overline{x_2} - x_2) Q_{1,2}^{m-1,0} - \overline{a_2}\delta_{m-1}) \\ & + c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} - c_2 tv(\overline{x_2} - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\ =& tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) + c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} - c_2 tv(\overline{x_2} - x_2) Q_{1,2}^{m+1,0}) - \overline{a_2} tv\delta_{m-1}) \end{aligned}$$

Combining this with equation (15) leads to

$$0 = tv(R^* Q_1^{m+1} - R Q_1^{m+1,*}) - c_2 tv(\overline{x_2} - x_2) Q_{1,2}^{m+1,0} - \overline{a_2} tv \delta_{m-1} + \overline{a_2} \delta_m$$
$$0 = R^* Q_1^{m+1} - R Q_1^{m+1,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m+1,0} + \overline{a_2} \delta_{m+1}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1}$$

This function δ_{m+1} is also not dependent on a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$(c_2 \overline{x_2} - \overline{a_2} x_2) Q_1^m(x_2) - (c_2 x_2 - \overline{a_2 x_2}) Q_1^m(\overline{x_2}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2} \delta_m = 0$$

We start by extracting the $[x_2^1]$ coefficient

$$c_2 Q_{1,2}^{m,2} - \overline{a_2} Q_{1,2}^{m,0} - c_2 Q_{1,2}^{m,0} + c_2 Q_{1,2}^{m,0} + [x_2^1] \overline{a_2} \delta_m = 0$$
$$c_2 Q_{1,2}^{m,2} - \overline{a_2} Q_{1,2}^{m,0} + [x_2^1] \overline{a_2} \delta_m = 0$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geqslant 2$

$$c_2 Q_{1,2}^{m,n+2} - \overline{a_2} Q_{1,2}^{m,n} + [x_2^{n+1}] \overline{a_2} \delta_m = 0$$

This formula also contains the $n = 0$ case so we treat them together. We can set $a_2 = 1$ to get

$$-\breve{Q}_{1,2}^{m,n} + [x_2^{n+1}] \delta_m = 0$$

Substituting this back into the above equation gives

$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 6. For integers $m, n \geqslant 0$ we have the following relations

$$b_1 Q_{1,2}^{m+2,n} = Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n}$$
$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}$$

We can sum up those equations to obtain functional equations

Corollary 7. For integers $m \geqslant 0$ we have the following relations

$$Q_1^m - \widehat{Q}_1^m = b_1 Q_1^{m+2}$$
$$\breve{Q}_1^m - \breve{\widehat{Q}}_1^m = b_1 \breve{Q}_1^{m+2}$$
$$x_1^2(Q_2^n - \widehat{Q}_2^n) = b_1(Q_2^n - Q_{1,2}^{1,n} x_1 - Q_{1,2}^{0,n})$$
$$x_1^2(\breve{Q}_2^n - \breve{\widehat{Q}}_2^n) = b_1(\breve{Q}_2^n - \breve{Q}_{1,2}^{1,n} x_1 - \breve{Q}_{1,2}^{0,n})$$
$$x_1^2(Q - \widehat{Q}) = b_1(Q - x_1 Q_1^1 - Q_1^0)$$
$$x_1^2(\breve{Q} - \breve{\widehat{Q}}) = b_1(\breve{Q} - x_1 \breve{Q}_1^1 - \breve{Q}_1^0)$$

Corollary 8. For integers $n \geqslant 0$ we have the following relations

$$\begin{aligned}
Q_2^n - \breve{Q}_2^n &= b_2 Q_2^{n+2} \\
\widehat{Q}_2^n - \breve{\widehat{Q}}_2^n &= b_2 \widehat{Q}_2^{n+2} \\
x_2^2(Q_1^m - \breve{Q}_1^m) &= b_2(Q_1^m - Q_{1,2}^{m,1} x_1 - Q_{1,2}^{m,0}) \\
x_2^2(\widehat{Q}_1^m - \breve{\widehat{Q}}_1^m) &= b_2(\widehat{Q}_1^m - \widehat{Q}_{1,2}^{m,1} x_1 - \widehat{Q}_{1,2}^{m,0}) \\
x_2^2(Q - \breve{Q}) &= b_2(Q - x_2 Q_2^1 - Q_2^0) \\
x_2^2(\widehat{Q} - \breve{\widehat{Q}}) &= b_2(\widehat{Q} - x_2 \widehat{Q}_2^1 - \widehat{Q}_2^0)
\end{aligned}$$

Every other equation is obtained from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. Here we can hope to set some telescoping sum type of thing in order to say more about $Q_{1,2}^{0,0}$

5 A little aside

If we could prove the validity of one of the functional equations in the corollaries (5th row in each corollary) we would then be able to extract coefficients and obtain the formulas in the first corollary. However, since we have a direct proof already, this section is kind of irrelevant now. We'll just keep it for now because it has some interesting math. The steps from now on are almost exactly the same as those that you'd have to do to prove the Dyck path formula. We'll prove the last one involving a b_1 coefficient. First we have

$$\begin{aligned}
K &= 1 - tS = 1 - t(x_1 + \overline{x_1})(x_2 + \overline{x_2}) \\
K_1 &= c_1 - t\overline{x_1} S_1^{-1} = c_1 + \widehat{K}_1 \\
K_2 &= c_2 - t\overline{x_2} S_2^{-1} \\
K_{1,2} &= c_1 c_2 - t\overline{x_1 x_2} = c_1 c_2 + \widehat{K}_{1,2}
\end{aligned}$$

We also need the coefficient extractions

$$\begin{aligned}
[x_1^0]: &\quad Q_1^0 - c_2 Q_{1,2}^{0,0} - t a_1 S_1^{-1} Q_1^1 = \overline{a_2} \\
[x_1^0]: &Q_1^1 - t S_1^{-1} Q_1^2 - t S_1^{-1} Q_1^0 + t\overline{x_2} Q_{1,2}^{2,0} + t\overline{x_2} Q_{1,2}^{0,0} = 0 \\
&- t\overline{x_1} S_1^{-1} Q_1^2 + t\overline{x_1 x_2} Q_{1,2}^{2,0} = t\overline{x_1} S_1^{-1} Q_1^0 - t\overline{x_1 x_2} Q_{1,2}^{0,0} - \overline{x_1} Q_1^1
\end{aligned}$$

which we'll use soon. We first start with

$$\begin{aligned}
KQ &= K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \overline{a_1 a_2} \\
&= (c_1 + \widehat{K}_1) Q_1^0 + K_2 Q_2^0 - (c_1 c_2 + \widehat{K}_{1,2}) Q_{1,2}^{0,0} + \overline{a_1 a_2} \\
K\widehat{Q} &= \widehat{K}_1 \widehat{Q}_1^0 + K_2 \widehat{Q}_2^0 - \widehat{K}_{1,2} \widehat{Q}_{1,2}^{0,0} + \overline{a_2}
\end{aligned}$$

Then we have

$$\begin{aligned}
KQ - K\widehat{Q} &= \widehat{K}_1(Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2(Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2}(Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + (c_1 Q_1^0 - c_1 c_2 Q_{1,2}^{0,0} + \overline{a_1 a_2} - \overline{a_2}) \\
&= \widehat{K}_1(Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2(Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2}(Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + t b_1 S_1^{-1} Q_1^1
\end{aligned}$$

Also

$$\begin{aligned}\widehat{K}_1(Q_1^0-\widehat{Q}_1^0)-\widehat{K}_{1,2}(Q_{1,2}^{0,0}-\widehat{Q}_{1,2}^{0,0}) =&\widehat{K}_1 b_1 Q_1^2-\widehat{K}_{1,2}b_1 Q_{1,2}^{2,0}\\ =&b_1(\widehat{K}_1 Q_1^2-\widehat{K}_{1,2}Q_{1,2}^{0,0}\\ =&b_1(-t\overline{x_1}S_1^{-1}Q_1^2+t\overline{x_1x_2}Q_{1,2}^{2,0})\\ =&b_1(t\overline{x_1}S_1^{-1}Q_1^0-t\overline{x_1x_2}Q_{1,2}^{0,0}-\overline{x_1}Q_1^1)\end{aligned}$$

Next

$$\begin{aligned}K_2(Q_2^0-\widehat{Q}_2^0) &= (c_2-t\overline{x_1}^2S_2^{-1})\overline{x_2}b_1(Q_2^0-Q_{1,2}^{0,0})\\ &= -t\overline{x_1}^2\overline{x_2}b_1S_2^{-1}(Q_2^0-Q_{1,2}^{0,0})+c_2b_1\overline{x_1}^2(Q_2^0-Q_{1,2}^{0,0})\end{aligned}$$

We can finally combine these expressions into

$$x_1^2(KQ-K\widehat{Q}) = b_1x_1^2(tS_1^{-1}Q_1^1+t\overline{x_1}S_1^{-1}Q_1^0-t\overline{x_1x_2}Q_{1,2}^{0,0}-\overline{x_1}Q_1^1-t\overline{x_1}^2\overline{x_2}S_2^{-1}(Q_2^0-Q_{1,2}^{0,0})+c_2\overline{x_1}^2(Q_2^0-Q_{1,2}^{0,0})$$

We cancel out the b_1 and we have the equation

$$KQ-K(x_1Q_1^1+Q_1^0)) = tx_1S_1^{-1}(x_1Q_1^1+Q_1^0)-(tx_1\overline{x_2}Q_{1,2}^{0,0}+x_1Q_1^1+t\overline{x_2}S_2^{-1}(Q_2^0-Q_{1,2}^{0,0}))-c_2(Q_2-Q_{1,2}^{0,0})$$

Note that

$$K+tx_1S_1^{-1} = 1-t\overline{x_1}S_1^{-1}$$

Once we expand the KQ term on the LHS we get

$$\begin{aligned}&c_1Q_1^0-t\overline{x_1}S_1^{-1}Q_1^0+c_2Q_2^0-t\overline{x_2}S_1^{-1}Q_2^0+c_1c_2Q_{1,2}^{0,0}+t\overline{x_1x_2}Q_{1,2}^{0,0}+\overline{a_1a_2}\\ =&(1-t\overline{x_1}S_1^{-1})(x_1Q_1^1+Q_1^0)-x_1Q_1^1+t\overline{x_1x_2}Q_{1,2}^{0,0}-t\overline{x_2}S_2^{-1}Q_2^0+c_2(Q_2^0-Q_{1,2}^{0,0})\end{aligned}$$

and after careful book-keeping all the terms cancel out one by one. This completes the long-winded proof of the identity.